

The Making of Vehicle Currencies

Evidence from DeFi Route-Level Markets

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Motivation: vehicle currencies are routing infrastructure

- A vehicle currency is not just a large asset or a popular endpoint.
- It is an asset that other trades route through when direct exchange is missing, thin, or costly.
- Classic foreign-exchange markets make the vehicle role hard to observe directly.
- DeFi gives transaction-level routes, liquidity, endpoint tokens, and settlement traces.

Research question

When and why does an asset become the intermediary through which other assets trade?

One-sentence thesis

Main claim

Vehicle currencies are made by route feasibility and liquidity concentration: they protect traders from missing or thin direct markets, persist with vehicle-linked liquidity, rotate under stress, and become more virtual when settlement architecture nets physical token movement.

- The paper is not a WETH-only story.
- WETH, stablecoins, V3, and V4 are empirical test beds for general vehicle-currency mechanisms.

Why DeFi is a useful laboratory

What traditional data often hide

- route choice
- intermediate asset identity
- direct-route availability
- execution-cost counterfactuals
- settlement implementation

What the DeFi data reveal

- reconstructed multi-hop routes
- exact source, destination, vehicle token
- pool liquidity and LP allocation
- route-cost quotes and no-direct cases
- receipt-level transfer incidence

Measurement: separate endpoint demand from bridge use

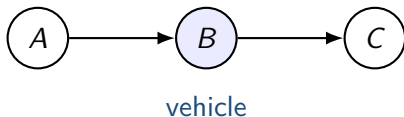
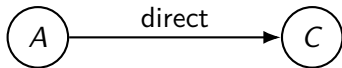
Main vehicle-use proxy

$$\text{BridgeShare}_{k,t} = \frac{\text{USD volume of indirect routes where } k \text{ is intermediate}}{\text{USD volume of all indirect routes}}$$

- Main object is conditional on indirect routing.
- All-route bridge share reports economic scope.
- Endpoint share is a control/diagnostic, not the vehicle proxy.

Evidence packet Table 1 / Table 2.

Route objects



A token is counted as a vehicle only if it is an intermediate, not the source or destination.

Empirical spine

1. **Formation:** direct-market incompleteness and thin-direct protection.
Evidence: route availability, depth, and cost advantage predict vehicle use.
2. **Liquidity:** liquidity and route use persist together.
Evidence: LP concentration predicts BridgeShare; lagged BridgeShare predicts future LP.
3. **Stress:** risky incumbent loses share on impact.
Evidence: WETH-minus-stable share falls on stress days.
4. **Architecture:** direct-market deepening reduces vehicle dependence.
Evidence: post-V3 no-direct/WETH cases fall.
5. **Settlement:** netting separates route use from physical movement.
Evidence: V4 lowers intermediary-token transfer incidence.

Proposition 1: direct-market incompleteness

Statement

For an endpoint pair, a candidate vehicle expands the feasible execution set when the direct route is unavailable or thin. Conditional on both routes being feasible, the vehicle route improves execution only when its route cost is below the direct-route cost.

Empirical margins

- no-direct, vehicle-available cases
- thin direct route, vehicle route available
- common-support cost advantage
- vehicle-route depth

Result 1: formation loads on route opportunity

	Actual vehicle share	Vehicle share $t + 7$	Vehicle share $t + 30$
Route-cost advantage, per 100 bp	0.1462*** ($p < 0.001$)	0.0001*** ($p = 0.002$)	0.0002*** ($p < 0.001$)
Vehicle-route availability	19.1385*** ($p < 0.001$)	0.0940*** ($p < 0.001$)	0.1254*** ($p < 0.001$)
Vehicle-route depth	28.8502*** ($p < 0.001$)		
LP concentration		0.1117*** ($p < 0.001$)	
Lagged vehicle share			0.3630*** ($p < 0.001$)
FE	pair $\times$ date	token, date	token, date
Observations	189,143	9,380	9,265

Evidence packet Table 4.

Result 1 interpretation

- The vehicle role is not simply “the cheapest route everywhere.”
- The strongest interpretation is **availability and thin-direct protection**.
- Common-support cost advantage is one mechanism, but not the whole vehicle value.
- This framing is important for not making a false universal-cheaper-route claim.

Proposition 2: liquidity-route persistence

Statement

Vehicle-linked liquidity and vehicle use are jointly persistent: liquidity around a candidate vehicle predicts future vehicle use, and current vehicle use predicts future vehicle-linked liquidity.

- This is a reduced-form persistence claim.
- It is not yet an exogenous LP-shock causal design.

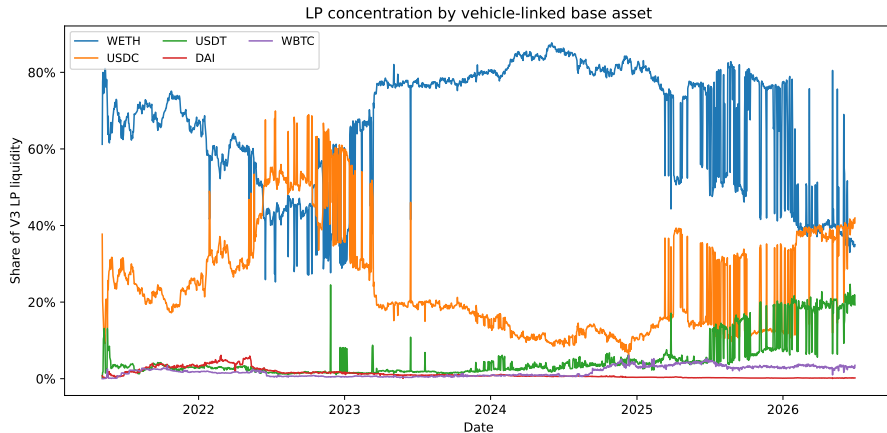
Result 2: liquidity and route use predict each other

	Vehicle share $t + 7$	LP concentration $t + 7$	Log LP liquidity $t + 7$
LP concentration	0.1117*** ($p < 0.001$)		
Lagged vehicle share		0.0419*** ($p < 0.001$)	1.1713*** ($p < 0.001$)
FE	token, date	token, date	token, date
SE clustering	date	date	date
Observations	9,380	9,380	9,380

- Liquidity concentration predicts future vehicle use.
- Vehicle use predicts future vehicle-linked liquidity.

Evidence packet Table 5.

Figure: vehicle shares and liquidity



Generated exhibit: LP concentration time series.

Proposition 3: impact stress rotation

Statement

A risk or credibility shock to an incumbent vehicle reduces its route-intermediation share on impact relative to safer substitutes, conditional on common route opportunities.

- Model is interval-neutral: it predicts an impact response.
- Empirically strongest at the same-day event window.
- Hourly/weekly results are useful duration/noise diagnostics, not the headline.

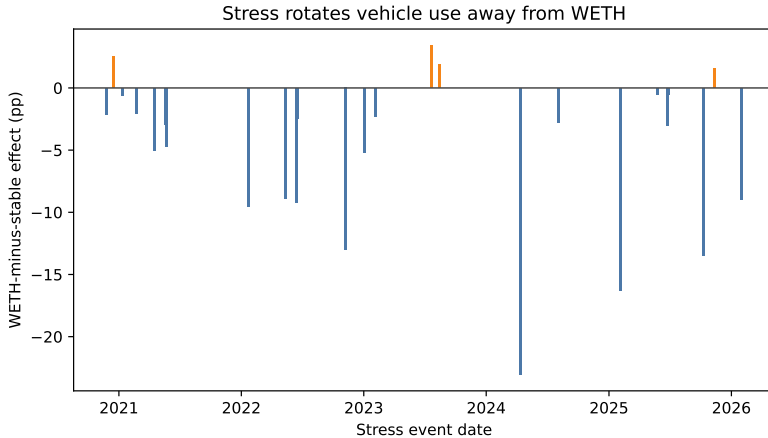
Result 3: stress rotates away from WETH

	WETH share	Stable share	WETH-stable gap	Direct share	Indirect volume
Stress event day	−1.48** pp ($p = 0.018$)	1.48** pp ($p = 0.018$)	−2.96** pp ($p = 0.018$)	−0.25 pp ($p = 0.679$)	0.51*** log pts ($p = 0.005$)

- Same-day common-support rotation is the cleanest stress result.
- Threshold and non-overlap checks remain negative and statistically significant.

Evidence packet Table 6.

Figure: common-support stress rotation



Generated exhibit: stress common-support vehicle rotation.

Proposition 4a: direct-market-deepening architecture

Statement

An architecture that increases direct pairwise depth reduces reliance on vehicle routes that exist only because the direct route is absent or thin.

- Empirical test bed: concentrated-liquidity launch.
- Defensible claim: no-direct/WETH-available cases fall.
- Do not claim every V3 outcome has clean causal timing.

Result 4a: V3 reduces no-direct dependence

	No-direct WETH	Q1 direct	Q1 no-direct	Q2 no-direct
Post V3	−25.81*** ($p < 0.001$)	56.42*** ($p < 0.001$)	−33.03*** ($p = 0.007$)	−47.51*** ($p < 0.001$)
FE	pair	pair	pair	pair
SE clustering	pair	pair	pair	pair

- Strongest message: architecture can reduce vehicle dependence by expanding direct-route opportunity.

Evidence packet Table 7.

Proposition 4b: settlement netting

Statement

Settlement netting can preserve the route-pricing role of a vehicle while reducing physical transfers of the intermediary token.

- Empirical test bed: matched V3 and V4 route units.
- Route vehicle use and physical settlement movement become separable objects.

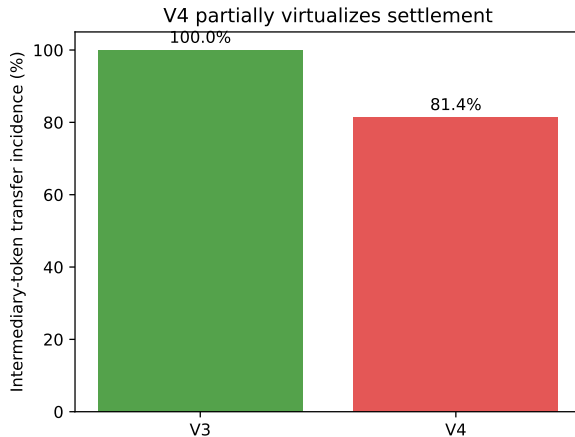
Result 4b: V4 virtualizes part of the vehicle role

	All sizes	Small	Medium	Large
V4 - V3 transfer incidence	-18.6*** pp ($p < 0.001$)	-39.4*** pp ($p < 0.001$)	-12.3*** pp ($p = 0.002$)	-3.2* pp ($p = 0.083$)
V3 transfer incidence	100.0%	100.0%	100.0%	100.0%
V4 transfer incidence	81.4%	60.6%	87.7%	96.8%

- V4 does not eliminate vehicle currencies.
- It makes part of the vehicle role virtual: the token can remain the route intermediary without moving physically in every route unit.

Evidence packet Table 8.

Figure: settlement-transfer incidence



Generated exhibit: V4 settlement transfer incidence.

Additional mechanism: common vehicle-linked liquidity

	Full	Stress	High veh.	Low veh.	Excl. top
Market factor	0.7063*** ($p < 0.001$)	0.6987*** ($p < 0.001$)	0.6572*** ($p < 0.001$)	0.5276*** ($p < 0.001$)	0.7062*** ($p < 0.001$)
Vehicle factor	0.1626*** ($p < 0.001$)	0.1586*** ($p < 0.001$)	0.2174*** ($p < 0.001$)	-0.0921 ($p = 0.307$)	0.1670*** ($p < 0.001$)
Vehicle factor $\times$ stress		0.1064** ($p = 0.030$)			

- Same-vehicle pools share a liquidity component beyond market-wide liquidity.
- This can be a main slide or a backup, depending on time.

Evidence packet Table 9.

What the evidence supports

1. **Formation:** route opportunity and depth predict vehicle use.
2. **Persistence:** vehicle use and vehicle-linked liquidity are persistent together.
3. **Rotation:** stress moves flow away from risky incumbent vehicle use toward stable substitutes on impact.
4. **Architecture:** market design changes direct-route feasibility.
5. **Settlement:** netting separates route intermediation from physical token movement.

What the evidence does not claim

- Not: WETH is always the cheapest route.
- Not: liquidity feedback is causally identified from an exogenous LP shock.
- Not: stress rotation is a permanent regime shift in every window.
- Not: V4 removes vehicle currencies.
- Not: Curve/Fluid are exact-quoted in the main route-cost panel.

Paper discipline

The strongest contribution comes from bounded claims that match the actual empirical objects.

Contribution

- **Vehicle-currency literature:** measures vehicle use directly as route intermediation.
- **Liquidity/intermediation literature:** shows vehicle status is tied to liquidity concentration and route opportunity.
- **Market design:** shows architecture changes vehicle reliance and settlement implementation.
- **DeFi evidence:** turns transparent route-level ledgers into a laboratory for classic monetary intermediation questions.

Next decisions for the Nanyang talk

- How technical should the model section be for this audience?
- Should common liquidity be main-deck or backup?
- How much DeFi institutional detail is needed before the results?
- Which result should be the punchline: liquidity formation, stress rotation, or V4 virtualization?

My recommendation

Lead with measurement and formation, use stress and V4 as sharp payoffs, keep common liquidity as backup unless the audience is strongly market-microstructure focused.

Conclusion

Takeaway

Vehicle currencies are not just dominant assets. They are route-level liquidity institutions whose role depends on direct-market incompleteness, liquidity concentration, stress-state substitution, and settlement architecture.